

TASK 12: Log Monitoring & Analysis

1) Understand Log Types.

Log File	Purpose
/var/log/auth.log	Login attempts, sudo usage
/var/log/syslog	System activity
/var/log/kern.log	Kernel events
/var/log/ufw.log	Firewall logs (if enabled)

2) Analyze Authentication Logs

View Authentication Logs:

```
(hari@ vbox) - [~]
$ sudo journalctl
Jan 20 00:01:09 vbox kernel: Linux version 6.12.25-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-14 (Debian 14.2.0-
Jan 20 00:01:09 vbox kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.25-amd64 root=UUID=ad862d53-2c37-4375-977
Jan 20 00:01:09 vbox kernel: BIOS-provided physical RAM map:
Jan 20 00:01:09 vbox kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
Jan 20 00:01:09 vbox kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
Jan 20 00:01:09 vbox kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
Jan 20 00:01:09 vbox kernel: BIOS-e820: [mem 0x0000000000100000-0x000000000009e9ffff] usable
Jan 20 00:01:09 vbox kernel: BIOS-e820: [mem 0x000000000009f0000-0x000000000009e9ffff] ACPI data
Jan 20 00:01:09 vbox kernel: BIOS-e820: [mem 0x000000000fec00000-0x000000000fec00fff] reserved
Jan 20 00:01:09 vbox kernel: BIOS-e820: [mem 0x000000000fee00000-0x000000000fee00fff] reserved
Jan 20 00:01:09 vbox kernel: BIOS-e820: [mem 0x000000000fff00000-0x000000000fff00fff] reserved
Jan 20 00:01:09 vbox kernel: NX (Execute Disable) protection: active
Jan 20 00:01:09 vbox kernel: APIC: Static calls initialized
Jan 20 00:01:09 vbox kernel: SMBIOS 2.5 present.
Jan 20 00:01:09 vbox kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
Jan 20 00:01:09 vbox kernel: DMI: Memory slots populated: 0/0
Jan 20 00:01:09 vbox kernel: Hypervisor detected: KVM
Jan 20 00:01:09 vbox kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
Jan 20 00:01:09 vbox kernel: kvm-clock: using sched offset of 3169043179408 cycles
Jan 20 00:01:09 vbox kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_
Jan 20 00:01:09 vbox kernel: tsc: Detected 2995.198 MHz processor
Jan 20 00:01:09 vbox kernel: e820: update [mem 0x00000000-0x00000fff] usable => reserved
Jan 20 00:01:09 vbox kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
Jan 20 00:01:09 vbox kernel: last_pfn = 0x9ea00 max_arch_pfn = 0x400000000
Jan 20 00:01:09 vbox kernel: MTRR map: 3 entries (3 fixed + 0 variable; max 19), built from 8 variable MTRRs
Jan 20 00:01:09 vbox kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
Jan 20 00:01:09 vbox kernel: CPU MTRRs all blank - virtualized system.
Jan 20 00:01:09 vbox kernel: found SMP MP-table at [mem 0x0009fbf0-0x0009fbff]
Jan 20 00:01:09 vbox kernel: RAMDISK: [mem 0x28409000-0x301fbfff]
Jan 20 00:01:09 vbox kernel: ACPI: Early table checksum verification disabled
Jan 20 00:01:09 vbox kernel: ACPI: RSDP 0x00000000000E0000 000024 (v02 VBOX )
Jan 20 00:01:09 vbox kernel: ACPI: XSDT 0x0000000009E9F030 00003C (v01 VBOX VBOXXSDT 00000001 ASL 00000061)
Jan 20 00:01:09 vbox kernel: ACPI: FACP 0x0000000009E9F0F0 0000F4 (v04 VBOX VBOXFACP 00000001 ASL 00000061)
Jan 20 00:01:09 vbox kernel: ACPI: DSDT 0x0000000009E9F0610 002353 (v02 VBOX VBOXBIOS 00000002 INTL 20100528)
Jan 20 00:01:09 vbox kernel: ACPI: FACS 0x0000000009E9F0200 000040
Jan 20 00:01:09 vbox kernel: ACPI: APIC 0x0000000009E9F0240 00005C (v02 VBOX VBOXAPIC 00000001 ASL 00000061)
Jan 20 00:01:09 vbox kernel: ACPI: SSDT 0x0000000009E9F02A0 00036C (v01 VBOX VBOXCPUPT 00000002 INTL 20100528)
Jan 20 00:01:09 vbox kernel: ACPI: Reserving FACP table memory at [mem 0x9e9f00f0-0x9e9f01e3]
Jan 20 00:01:09 vbox kernel: ACPI: Reserving DSDT table memory at [mem 0x9e9f0610-0x9e9f2962]
Jan 20 00:01:09 vbox kernel: ACPI: FACP 0x0000000009E9F0F0 0000F4 (v04 VBOX VBOXFACP 00000001 ASL 00000061)
Jan 20 00:01:09 vbox kernel: ACPI: DSDT 0x0000000009E9F0610 002353 (v02 VBOX VBOXBIOS 00000002 INTL 20100528)
Jan 20 00:01:09 vbox kernel: ACPI: FACS 0x0000000009E9F0200 000040
Jan 20 00:01:09 vbox kernel: ACPI: APIC 0x0000000009E9F0240 00005C (v02 VBOX VBOXAPIC 00000001 ASL 00000061)
Jan 20 00:01:09 vbox kernel: ACPI: SSDT 0x0000000009E9F02A0 00036C (v01 VBOX VBOXCPUPT 00000002 INTL 20100528)
Jan 20 00:01:09 vbox kernel: ACPI: Reserving FACP table memory at [mem 0x9e9f00f0-0x9e9f01e3]
Jan 20 00:01:09 vbox kernel: ACPI: Reserving DSDT table memory at [mem 0x9e9f0610-0x9e9f2962]
Jan 20 00:01:09 vbox kernel: ACPI: Reserving FACS table memory at [mem 0x9e9f0200-0x9e9f023f]
Jan 20 00:01:09 vbox kernel: ACPI: Reserving FACS table memory at [mem 0x9e9f0200-0x9e9f023f]
Jan 20 00:01:09 vbox kernel: ACPI: Reserving APIC table memory at [mem 0x9e9f0240-0x9e9f029b]
Jan 20 00:01:09 vbox kernel: ACPI: Reserving SSDT table memory at [mem 0x9e9f02a0-0x9e9f060b]
Jan 20 00:01:09 vbox kernel: No NUMA configuration found
Jan 20 00:01:09 vbox kernel: Faking a node at [mem 0x0000000000000000-0x0000000009e9fffff]
Jan 20 00:01:09 vbox kernel: NODE_DATA(0) allocated [mem 0x9e9c5680-0x9e9efffff]
Jan 20 00:01:09 vbox kernel: Zone ranges:
Jan 20 00:01:09 vbox kernel: DMA [mem 0x0000000000001000-0x000000000009fffff]
Jan 20 00:01:09 vbox kernel: DMA32 [mem 0x0000000000100000-0x0000000009e9fffff]
Jan 20 00:01:09 vbox kernel: Normal empty
Jan 20 00:01:09 vbox kernel: Device empty
Jan 20 00:01:09 vbox kernel: Movable zone start for each node
Jan 20 00:01:09 vbox kernel: Early memory node ranges
```

Shows all system logs

Includes login, sudo, ssh, system events

View Recent Authentication Activity:

```

(hari@vbox)-[~]
$ sudo journalctl -xe
Feb 09 05:27:59 vbox kernel: 10:27:59.955181 X11 events Sending monitor positions (8 of them) to the host: VINF_
Feb 09 05:27:59 vbox kernel: 10:27:59.970700 X11 events received X11 event (89)
Feb 09 05:28:00 vbox kernel: 10:28:00.001287 X11 events RRSscreenChangeNotify event received
Feb 09 05:28:00 vbox kernel: 10:28:00.102757 dp-vsga-x11 output[0] successfully configured
Feb 09 05:28:00 vbox kernel: 10:28:00.177848 X11 events Monitor 0 (w,h)=(956,926) (x,y)=(0,0)
Feb 09 05:28:00 vbox kernel: 10:28:00.181653 X11 events Sending monitor positions (8 of them) to the host: VINF_
Feb 09 05:28:00 vbox kernel: 10:28:00.185239 X11 events received X11 event (89)
Feb 09 05:28:00 vbox kernel: 10:28:00.216297 X11 events RRSscreenChangeNotify event received
Feb 09 05:28:00 vbox kernel: 10:28:00.226111 X11 events Monitor 0 (w,h)=(956,926) (x,y)=(0,0)
Feb 09 05:28:00 vbox kernel: 10:28:00.226392 X11 events Sending monitor positions (8 of them) to the host: VINF_
Feb 09 05:28:00 vbox kernel: 10:28:00.226885 X11 events received X11 event (89)
Feb 09 05:28:00 vbox kernel: 10:28:00.227624 X11 events RRSscreenChangeNotify event received
Feb 09 05:28:00 vbox kernel: 10:28:00.295813 X11 events Monitor 0 (w,h)=(956,926) (x,y)=(0,0)
Feb 09 05:28:00 vbox kernel: 10:28:00.296070 X11 events Sending monitor positions (8 of them) to the host: VINF_
Feb 09 05:28:00 vbox kernel: 10:28:00.296291 X11 events received X11 event (89)
Feb 09 05:28:00 vbox kernel: 10:28:00.296441 X11 events RRSscreenChangeNotify event received
Feb 09 05:28:00 vbox kernel: 10:28:00.297089 X11 events Monitor 0 (w,h)=(956,926) (x,y)=(0,0)
Feb 09 05:28:00 vbox kernel: 10:28:00.302924 X11 events Sending monitor positions (8 of them) to the host: VINF_
Feb 09 05:28:00 vbox kernel: 10:28:00.623432 dp-vsga-x11 output[0] successfully configured
Feb 09 05:28:00 vbox kernel: 10:28:00.805695 X11 events received X11 event (89)
Feb 09 05:28:00 vbox kernel: 10:28:00.851419 X11 events RRSscreenChangeNotify event received
Feb 09 05:28:00 vbox kernel: 10:28:00.884298 X11 events Monitor 0 (w,h)=(955,925) (x,y)=(0,0)
Feb 09 05:28:00 vbox kernel: 10:28:00.887754 X11 events Sending monitor positions (8 of them) to the host: VINF_
Feb 09 05:28:00 vbox kernel: 10:28:00.973256 X11 events received X11 event (89)
Feb 09 05:28:00 vbox kernel: 10:28:00.973479 X11 events RRSscreenChangeNotify event received
Feb 09 05:28:00 vbox kernel: 10:28:00.974008 X11 events Monitor 0 (w,h)=(955,925) (x,y)=(0,0)
Feb 09 05:28:00 vbox kernel: 10:28:00.974269 X11 events Sending monitor positions (8 of them) to the host: VINF_
Feb 09 05:28:00 vbox kernel: 10:28:00.974897 X11 events received X11 event (89)
Feb 09 05:28:00 vbox kernel: 10:28:00.975077 X11 events RRSscreenChangeNotify event received
Feb 09 05:28:00 vbox kernel: 10:28:00.975990 X11 events Monitor 0 (w,h)=(955,925) (x,y)=(0,0)
Feb 09 05:28:00 vbox kernel: 10:28:00.976203 X11 events Sending monitor positions (8 of them) to the host: VINF_
Feb 09 05:28:00 vbox kernel: 10:28:00.976483 X11 events received X11 event (89)
Feb 09 05:28:00 vbox kernel: 10:28:00.976628 X11 events RRSscreenChangeNotify event received
Feb 09 05:28:00 vbox kernel: 10:28:00.977109 X11 events Monitor 0 (w,h)=(955,925) (x,y)=(0,0)
Feb 09 05:28:00 vbox kernel: 10:28:00.977613 X11 events Sending monitor positions (8 of them) to the host: VINF_
Feb 09 05:28:24 vbox sudo[186437]: hari : TTY=pts/0 ; PWD=/home/hari ; USER=root ; COMMAND=/usr/bin/cat /var/log
Feb 09 05:28:24 vbox sudo[186437]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Feb 09 05:28:24 vbox sudo[186437]: pam_unix(sudo:session): session closed for user root
Feb 09 05:30:02 vbox sudo[187263]: hari : TTY=pts/0 ; PWD=/home/hari ; USER=root ; COMMAND=/usr/bin/journalctl
Feb 09 05:30:02 vbox sudo[187263]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Feb 09 05:30:02 vbox sudo[187263]: pam_unix(sudo:session): session closed for user root
Feb 09 05:30:37 vbox sudo[187552]: hari : TTY=pts/0 ; PWD=/home/hari ; USER=root ; COMMAND=/usr/bin/journalctl
Feb 09 05:30:37 vbox sudo[187552]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Feb 09 05:30:37 vbox sudo[187552]: pam_unix(sudo:session): session closed for user root
Feb 09 05:31:01 vbox sudo[187772]: hari : TTY=pts/0 ; PWD=/home/hari ; USER=root ; COMMAND=/usr/bin/journalctl
Feb 09 05:31:01 vbox sudo[187772]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Feb 09 05:31:02 vbox sudo[187772]: pam_unix(sudo:session): session closed for user root
Feb 09 05:32:17 vbox sudo[188387]: hari : TTY=pts/0 ; PWD=/home/hari ; USER=root ; COMMAND=/usr/bin/journalctl
Feb 09 05:32:17 vbox sudo[188387]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Feb 09 05:32:17 vbox sudo[188387]: pam_unix(sudo:session): session closed for user root

```

Recent events

Errors & warnings

3) Identify Failed Login Attempts:

```

(hari@vbox)-[~]
$ sudo journalctl | grep -i "failed"
Jan 20 00:01:09 vbox systemd[1]: Failed to read SMBIOS type #11 object 0, ignoring: Bad message
Jan 20 00:01:32 vbox wireplumber[743]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.Na
meHasNoOwner
Jan 20 01:05:15 vbox systemd[1]: Failed to read SMBIOS type #11 object 0, ignoring: Bad message
Jan 20 01:05:39 vbox wireplumber[772]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.Na
meHasNoOwner
Jan 20 01:05:50 vbox lightdm[740]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpo
int is not connected
Jan 20 01:05:50 vbox wireplumber[885]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.Na
meHasNoOwner
Jan 20 01:05:50 vbox lightdm[848]: Failed to write utmpx: No such file or directory
Jan 20 01:05:59 vbox colord[1232]: failed to get edid data: EDID length is too small
Jan 20 01:06:08 vbox dbus-daemon[543]: [system] Activation via systemd failed for unit 'dbus-org.freedesktop.Avahi.
service': Unit dbus-org.freedesktop.Avahi.service not found.
Jan 20 01:06:08 vbox dbus-daemon[543]: [system] Activation via systemd failed for unit 'dbus-org.freedesktop.Avahi.
service': Unit dbus-org.freedesktop.Avahi.service not found.
Jan 20 01:06:12 vbox dbus-daemon[543]: [system] Activation via systemd failed for unit 'dbus-org.bluez.service': Un
it dbus-org.bluez.service not found.
Jan 20 01:06:13 vbox dbus-daemon[881]: [session uid=1000 pid=881 pidfd=5] Activation via systemd failed for unit 'e
volution-source-registry.service': Unit evolution-source-registry.service not found.
Jan 20 01:06:13 vbox dbus-daemon[881]: [session uid=1000 pid=881 pidfd=5] Activation via systemd failed for unit 'e
volution-source-registry.service': Unit evolution-source-registry.service not found.
Jan 20 01:54:31 vbox systemd[1]: systemd-journald.service: Failed with result 'watchdog'.
Jan 20 01:56:11 vbox xfce4-screensaver-dialog[15147]: pam_unix(xfce4-screensaver:account): setuid failed: Operation
not permitted
Jan 20 02:06:56 vbox xfce4-screensaver-dialog[20264]: pam_unix(xfce4-screensaver:account): setuid failed: Operation
not permitted
Jan 20 02:40:23 vbox systemd[1]: systemd-journald.service: Failed with result 'watchdog'.
Jan 20 02:40:23 vbox systemd[1]: systemd-logind.service: Failed with result 'watchdog'.
Jan 20 02:41:49 vbox xfce4-screensaver-dialog[22563]: pam_unix(xfce4-screensaver:account): setuid failed: Operation
not permitted
Jan 20 02:41:49 vbox xfce4-screensaver-dialog[22563]: pam_unix(xfce4-screensaver:account): setuid failed: Operation
not permitted
Jan 20 02:41:49 vbox xfce4-screensaver-dialog[48109]: pam_unix(xfce4-screensaver:account): setuid failed: Operation
not permitted
Jan 20 04:21:59 vbox systemd[1]: systemd-udev.service: Failed with result 'watchdog'.
Jan 20 04:21:59 vbox systemd[1]: systemd-journald.service: Failed with result 'watchdog'.
Jan 20 04:22:35 vbox xfce4-screensaver-dialog[53271]: pam_unix(xfce4-screensaver:account): setuid failed: Operation
not permitted
Jan 20 04:34:03 vbox xfce4-screensaver-dialog[58756]: pam_unix(xfce4-screensaver:account): setuid failed: Operation
not permitted
Jan 20 04:34:19 vbox lightdm[848]: Failed to write utmpx: No such file or directory
Jan 20 04:34:19 vbox kernel: 09:34:19.207868 seamless Error: Failed to open X11 display
Jan 20 04:34:19 vbox systemd[859]: xfce4-notifdyd.service: Failed with result 'exit-code'.
Jan 20 04:34:19 vbox systemd[859]: xdg-desktop-portal-gtk.service: Failed with result 'exit-code'.
Jan 20 04:34:19 vbox kernel: 09:34:19.512041 RTHrdPP Error: Failed to open X11 display
Jan 20 04:34:19 vbox kernel: 09:34:19.514875 main Error: Failed to open X11 display
Jan 20 04:34:24 vbox dbus-daemon[543]: [system] Activation via systemd failed for unit 'dbus-org.freedesktop.nm-dis
patcher.service': Refusing activation, D-Bus is shutting down.
Jan 23 08:07:02 vbox systemd[1]: Failed to read SMBIOS type #11 object 0, ignoring: Bad message
Jan 23 08:07:33 vbox wireplumber[890]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.Na
meHasNoOwner
Jan 23 08:07:44 vbox lightdm[857]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpo
int is not connected
Jan 23 08:07:49 vbox lightdm[965]: Failed to write utmpx: No such file or directory
Jan 23 08:07:47 vbox wireplumber[997]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.Na
meHasNoOwner
Jan 23 08:07:58 vbox colord[1348]: failed to get edid data: EDID length is too small
Jan 23 08:08:08 vbox dbus-daemon[518]: [system] Activation via systemd failed for unit 'dbus-org.freedesktop.Avahi.
service': Unit dbus-org.freedesktop.Avahi.service not found.
Jan 23 08:08:08 vbox dbus-daemon[518]: [system] Activation via systemd failed for unit 'dbus-org.freedesktop.Avahi.
service': Unit dbus-org.freedesktop.Avahi.service not found.
Jan 23 08:08:11 vbox dbus-daemon[518]: [system] Activation via systemd failed for unit 'dbus-org.bluez.service': Un

```

Check Failed Login Attempts:

➔ If NO output appears:

No failed login attempts occurred recently

System authentication is currently secure

➔ Understand the Output:

Failed password for invalid user admin from 192.168.1.10 authentication failure.

Someone entered a **wrong password**

Possible brute-force or unauthorized attempt

4) Detect Anomalies

Check Privilege Escalation:

```
(hari@vbox)-[~]
└─$ sudo journalctl | grep sudo
Jan 20 01:06:31 vbox sudo[1801]: hari : TTY=pts/2 ; PWD=/home/hari ; USER=root ; COMMAND=/usr/bin/apt update
Jan 20 01:06:31 vbox sudo[1801]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Jan 20 01:06:36 vbox sudo[1801]: pam_unix(sudo:session): session closed for user root
Jan 20 02:59:28 vbox sudo[31238]: hari : TTY=pts/0 ; PWD=/home/hari/linux-security-demo ; USER=root ; COMMAND=/usr/bin/chown root:root file1.txt
Jan 20 02:59:28 vbox sudo[31238]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Jan 20 02:59:29 vbox sudo[31238]: pam_unix(sudo:session): session closed for user root
Jan 20 03:04:25 vbox sudo[33628]: hari : TTY=pts/0 ; PWD=/home/hari/linux-security-demo ; USER=root ; COMMAND=/usr/bin/apt install ufw
Jan 20 03:04:25 vbox sudo[33628]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Jan 20 03:05:01 vbox sudo[33628]: pam_unix(sudo:session): session closed for user root
Jan 20 03:12:19 vbox sudo[37747]: hari : TTY=pts/0 ; PWD=/home/hari/linux-security-demo ; USER=root ; COMMAND=/usr/sbin/ufw enable
Jan 20 03:12:21 vbox sudo[37747]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Jan 20 03:12:21 vbox sudo[37747]: pam_unix(sudo:session): session closed for user root
Jan 20 03:12:47 vbox sudo[38161]: hari : TTY=pts/0 ; PWD=/home/hari/linux-security-demo ; USER=root ; COMMAND=/usr/sbin/ufw status verbose
Jan 20 03:12:47 vbox sudo[38161]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Jan 20 03:12:48 vbox sudo[38161]: pam_unix(sudo:session): session closed for user root
Jan 20 03:13:15 vbox sudo[38432]: hari : TTY=pts/0 ; PWD=/home/hari/linux-security-demo ; USER=root ; COMMAND=/usr/sbin/ufw allow 22/tcp
Jan 20 03:13:15 vbox sudo[38432]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Jan 20 03:13:15 vbox sudo[38432]: pam_unix(sudo:session): session closed for user root
Jan 23 08:13:26 vbox sudo[4402]: hari : TTY=pts/0 ; PWD=/home/hari ; USER=root ; COMMAND=/usr/bin/apt update
Jan 23 08:13:26 vbox sudo[4402]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Jan 23 08:13:48 vbox sudo[4402]: pam_unix(sudo:session): session closed for user root
Jan 23 09:19:16 vbox sudo[5331]: hari : TTY=pts/0 ; PWD=/home/hari ; USER=root ; COMMAND=/usr/bin/apt install j
ohn -y
Jan 23 09:19:16 vbox sudo[5331]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Jan 23 09:27:05 vbox sudo[5331]: pam_unix(sudo:session): session closed for user root
Jan 26 11:20:24 vbox sudo[48140]: hari : TTY=pts/0 ; PWD=/home/hari ; USER=root ; COMMAND=/usr/bin/apt install
dvwa -y
Jan 26 11:20:24 vbox sudo[48140]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Jan 26 11:20:25 vbox sudo[48140]: pam_unix(sudo:session): session closed for user root
Jan 26 11:20:36 vbox sudo[48274]: hari : TTY=pts/0 ; PWD=/home/hari ; USER=root ; COMMAND=/usr/bin/apt install
dvwa -y
Jan 26 11:20:36 vbox sudo[48274]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Jan 26 11:22:48 vbox sudo[48274]: pam_unix(sudo:session): session closed for user root
Jan 26 11:22:58 vbox sudo[49402]: hari : TTY=pts/0 ; PWD=/home/hari ; USER=root ; COMMAND=/usr/bin/apt install
dvwa -y
Jan 26 11:22:58 vbox sudo[49402]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Jan 26 11:23:00 vbox sudo[49402]: pam_unix(sudo:session): session closed for user root
Jan 26 11:23:21 vbox sudo[49596]: hari : TTY=pts/0 ; PWD=/home/hari ; USER=root ; COMMAND=/usr/bin/apt install
dvwa -y
Jan 26 11:23:21 vbox sudo[49596]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Jan 26 11:23:23 vbox sudo[49596]: pam_unix(sudo:session): session closed for user root
Jan 26 11:23:35 vbox sudo[49716]: hari : TTY=pts/0 ; PWD=/home/hari ; USER=root ; COMMAND=/usr/bin/apt update
Jan 26 11:23:35 vbox sudo[49716]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Jan 26 11:36:56 vbox sudo[1855]: hari : TTY=pts/0 ; PWD=/home/hari ; USER=root ; COMMAND=/usr/bin/apt update
Jan 26 11:36:56 vbox sudo[1855]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Jan 26 11:48:01 vbox sudo[1855]: pam_unix(sudo:session): session closed for user root
Jan 26 11:48:35 vbox sudo[7570]: hari : TTY=pts/0 ; PWD=/home/hari ; USER=root ; COMMAND=/usr/bin/apt install d
vwa -y
Jan 26 11:48:35 vbox sudo[7570]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Jan 26 11:50:53 vbox sudo[7570]: pam_unix(sudo:session): session closed for user root
Jan 26 12:01:30 vbox sudo[23973]: hari : TTY=pts/0 ; PWD=/home/hari ; USER=root ; COMMAND=/usr/sbin/service apa
che2 start
Jan 26 12:01:30 vbox sudo[23973]: pam_unix(sudo:session): session opened for user root(uid=0) by hari(uid=1000)
Jan 26 12:01:32 vbox sudo[23973]: pam_unix(sudo:session): session closed for user root
Jan 26 12:01:51 vbox sudo[24181]: hari : TTY=pts/0 ; PWD=/home/hari ; USER=root ; COMMAND=/usr/sbin/service mys
ql start
```

User used **sudo**

Privilege escalation occurred

Normal admin activity

Identify Failed Login Events

Identify Firewall Block Events(UFW):

```
UFW BLOCK IN=eth0 SRC=10.0.2.15 DPT=21
```

- Firewall blocked a connection
- Possible network-level protection

What is SIEM? - Security Information and Event Management (SIEM) is a security solution that collects, stores, and analyzes log data from multiple systems such as servers, applications, firewalls, and network devices.

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Key Functions of SIEM:

Log Collection: Gathers logs from multiple sources in a centralized location

Event Correlation: Analyzes relationships between events to identify attack patterns

Alert Generation: Triggers alerts when suspicious or predefined conditions are met

Incident Investigation: Supports forensic analysis and reporting

Examples of SIEM Tools

- **Splunk** – Widely used for log analysis and security monitoring
- **ELK Stack (Elasticsearch, Logstash, Kibana)** – Open-source log analytics platform
- **IBM QRadar** – Enterprise SIEM for threat detection and compliance

7) Write Alerts

Alert Rule 1: Multiple Failed Login Attempts:

Condition:

- If 5 failed login attempts occur within 2 minutes

Reason:

- Indicates a possible brute-force attack

Alert Rule 2: Login from Unknown IP Address:

Condition:

- Login detected from an IP address not previously associated with the user

Reason:

- May indicate account compromise or unauthorized access